EnterpriseOS

Complete Master Feature Catalog

Enterprise SaaS / ERP / HRMS / CRM / Operations Platform

Technology baseline: Java Spring Boot | React + TypeScript | MySQL 8.4 LTS

Coverage: 21 functional domains | 316 proposed capabilities | 5 delivery phases

DOCUMENT STATUS: Proposed feature inventory only. Items are NOT automatically included in a contracted scope. Client must select Must/Should/Could, acceptance criteria, and deployment constraints before baseline approval.

1. Product Vision & Boundaries

A configurable multi-company business platform for company administration, employee lifecycle, operations, sales, procurement, finance and governance. Each tenant must be securely isolated. No AI, ML, LLM, or generative AI functionality is included.

2. Intended Users

- Platform users: Platform Owner, Platform Super Admin
- Tenant users: Company Owner, Company Admin, Department Manager, Employee, HR, Finance, Sales, Procurement, Project Manager, Auditor, Read-only Reviewer

3. Phase Overview

Phase	Delivery objective	Domains	Capabilities
1	Platform foundation and secure onboarding	4	56
2	People, attendance and project operations	4	62
3	Customer, stock, procurement and finance	5	77
4	Workflow, documents, notification and reporting	5	73
5	Commercial SaaS and production hardening	3	48

4. Feature Priority Legend

Must = proposed baseline; Should = recommended subsequent slice; Could = optional enhancement. All priorities are recommendations and may be changed during client workshops. “Phase” indicates proposed delivery wave, not a contractual delivery date.

5.1 Phase 1 Feature Specifications

Tenant & Company Onboarding

Feature ID	Feature / capability	Priority

F-101-01	Public company registration	Must
F-101-02	Email-domain verification	Must
F-101-03	Company profile and logo	Must
F-101-04	Company activation and suspension	Must
F-101-05	Company slug and unique identifiers	Must
F-101-06	Tenant-scoped application settings	Should
F-101-07	Company address and contact details	Should
F-101-08	Fiscal year and default currency	Should
F-101-09	Company timezone and locale	Should
F-101-10	Legal entity details	Should
F-101-11	Tenant lifecycle and deletion request	Should
F-101-12	Initial company admin provisioning	Could
F-101-13	Guided first-time setup checklist	Could
F-101-14	Company branch and location directory	Could

Requirement confirmation: confirm responsible user roles, validation rules, approval workflow, notifications, reporting needs and acceptance scenarios for every selected feature.

Identity, Authentication & Sessions

Feature ID	Feature / capability	Priority
F-102-01	Email/password login	Must
F-102-02	Secure password reset	Must
F-102-03	Email verification	Must
F-102-04	Password strength and history rules	Must
F-102-05	Multi-factor authentication (TOTP)	Must
F-102-06	Session and device listing	Should
F-102-07	Revoke individual or all sessions	Should
F-102-08	Account lockout and throttling	Should
F-102-09	Login attempt audit	Should
F-102-10	Refresh-token rotation	Should
F-102-11	Automatic session expiry	Should
F-102-12	Change password	Could
F-102-13	Passwordless invitation activation	Could
F-102-14	Deactivated user access revocation	Could

Requirement confirmation: confirm responsible user roles, validation rules, approval workflow, notifications, reporting needs and acceptance scenarios for every selected feature.

Roles, Permissions & Access Control

Feature ID	Feature / capability	Priority
F-103-01	System Super Admin role	Must
F-103-02	Tenant Owner role	Must
F-103-03	Company Admin role	Must
F-103-04	Department Manager role	Must
F-103-05	Employee role	Must
F-103-06	Custom role creation	Should
F-103-07	Permission catalog by module/action	Should
F-103-08	Role assignment and revocation	Should
F-103-09	Route and API authorization	Should
F-103-10	Department/branch scoped access	Should
F-103-11	Data ownership access restrictions	Should
F-103-12	Least-privilege defaults	Could
F-103-13	Privileged operation re-authentication	Could
F-103-14	Permission-change audit history	Could

Requirement confirmation: confirm responsible user roles, validation rules, approval workflow, notifications, reporting needs and acceptance scenarios for every selected feature.

Users, Organization & Configuration

Feature ID	Feature / capability	Priority
F-104-01	Invite users by email	Must
F-104-02	Bulk user import with validation	Must

F-104-03	User profile edit	Must
F-104-04	Activate/deactivate user	Must
F-104-05	Departments CRUD	Must
F-104-06	Reporting hierarchy	Should
F-104-07	Teams and job titles	Should
F-104-08	Office location management	Should
F-104-09	Employee directory basic view	Should
F-104-10	Organization chart basic view	Should
F-104-11	User search/filter/sort	Should
F-104-12	Timezone-aware date display	Could
F-104-13	Configurable numbering sequences	Could
F-104-14	Administrative landing dashboard	Could

Requirement confirmation: confirm responsible user roles, validation rules, approval workflow, notifications, reporting needs and acceptance scenarios for every selected feature.

5.2 Phase 2 Feature Specifications

Employee Lifecycle & HR Records

Feature ID	Feature / capability	Priority
F-205-01	Employee profile with custom fields	Must
F-205-02	Employee ID generation	Must
F-205-03	Employee onboarding checklist	Must
F-205-04	Emergency contacts	Must
F-205-05	Education and qualifications	Must
F-205-06	Work history	Should
F-205-07	Employment contracts and dates	Should
F-205-08	Probation tracking	Should
F-205-09	Promotion and transfer records	Should
F-205-10	Manager and department assignment	Should
F-205-11	Exit/resignation request	Should
F-205-12	Offboarding checklist	Could
F-205-13	Rehire employee	Could
F-205-14	Employee document expiry reminders	Could
F-205-15	HR-only confidential field access	Could

Requirement confirmation: confirm responsible user roles, validation rules, approval workflow, notifications, reporting needs and acceptance scenarios for every selected feature.

Attendance, Shifts & Timesheets

Feature ID	Feature / capability	Priority
F-206-01	Work shift templates	Must
F-206-02	Shift assignment and rotation	Must
F-206-03	Check-in/check-out capture	Must
F-206-04	Attendance correction requests	Must
F-206-05	Grace period and late rules	Must
F-206-06	Overtime policy	Should
F-206-07	Weekend and holiday calendars	Should
F-206-08	Remote/onsite attendance labels	Should
F-206-09	Attendance regularization approvals	Should
F-206-10	Attendance monthly summary	Should
F-206-11	Timesheet entries by project	Should
F-206-12	Timesheet approval	Could
F-206-13	Missing punch alerts	Could
F-206-14	Attendance export	Could
F-206-15	Cross-midnight shift handling	Could

Requirement confirmation: confirm responsible user roles, validation rules, approval workflow, notifications, reporting needs and acceptance scenarios for every selected feature.

Leave & Holiday Management

Feature ID	Feature / capability	Priority
F-207-01	Leave type and policy setup	Must
F-207-02	Accrual and carry-forward rules	Must
F-207-03	Leave balance ledger	Must
F-207-04	Leave request and withdrawal	Must
F-207-05	Calendar overlap checks	Must
F-207-06	Leave approval chains	Should
F-207-07	Holiday list by location	Should
F-207-08	Half-day and hourly leave	Should
F-207-09	Leave attachments	Should
F-207-10	Leave encashment eligibility policy	Should
F-207-11	Compensatory-off balance	Should
F-207-12	Manager team leave calendar	Could
F-207-13	Leave history and export	Could
F-207-14	Negative balance configuration	Could

Requirement confirmation: confirm responsible user roles, validation rules, approval workflow, notifications, reporting needs and acceptance scenarios for every selected feature.

Projects, Sprints & Task Management

Feature ID	Feature / capability	Priority
F-208-01	Project creation and archive	Must
F-208-02	Project members and roles	Must
F-208-03	Milestones and deliverables	Must
F-208-04	Task/subtask hierarchy	Must
F-208-05	Kanban workflow board	Must
F-208-06	Sprint/backlog management	Should
F-208-07	Task estimates and due dates	Should
F-208-08	Dependencies and blockers	Should
F-208-09	Status transition rules	Should
F-208-10	Task comments and mentions	Should
F-208-11	Attachments and links	Should
F-208-12	Issue priority and labels	Could
F-208-13	Assignee workload view	Could
F-208-14	Recurring tasks	Could
F-208-15	Project timeline/Gantt view	Could
F-208-16	Task activity history	Could
F-208-17	Timesheet-to-task linking	Could
F-208-18	Project risk register	Could

Requirement confirmation: confirm responsible user roles, validation rules, approval workflow, notifications, reporting needs and acceptance scenarios for every selected feature.

5.3 Phase 3 Feature Specifications

CRM & Sales Operations

Feature ID	Feature / capability	Priority
F-309-01	Lead intake and assignment	Must
F-309-02	Lead source tracking	Must
F-309-03	Lead status lifecycle	Must
F-309-04	Duplicate lead detection	Must
F-309-05	Lead conversion to account	Must
F-309-06	Contact and organization records	Should
F-309-07	Custom sales pipeline	Should
F-309-08	Deal stages and probability	Should
F-309-09	Opportunity owner and team	Should
F-309-10	Activity notes and follow-ups	Should
F-309-11	Sales quotes and revisions	Should

F-309-12	Quote approval	Could
F-309-13	Customer contracts and renewals	Could
F-309-14	Customer interaction history	Could
F-309-15	Lost-deal reason analysis	Could
F-309-16	Deal forecast by expected close date	Could
F-309-17	CRM access visibility rules	Could

Requirement confirmation: confirm responsible user roles, validation rules, approval workflow, notifications, reporting needs and acceptance scenarios for every selected feature.

Inventory & Warehousing

Feature ID	Feature / capability	Priority
F-310-01	Product and SKU catalog	Must
F-310-02	Product category and units	Must
F-310-03	Warehouses and storage locations	Must
F-310-04	Stock receipt and issue ledger	Must
F-310-05	Inter-warehouse transfer	Must
F-310-06	Low stock alerts	Should
F-310-07	Reorder points	Should
F-310-08	Stock adjustment approvals	Should
F-310-09	Batch/lot support	Should
F-310-10	Serial number tracking	Should
F-310-11	Stock reservation	Should
F-310-12	Stock valuation method policy	Could
F-310-13	Inventory cycle counts	Could
F-310-14	Inventory reconciliation	Could
F-310-15	Stock movement audit trail	Could
F-310-16	Supplier-linked item master	Could

Requirement confirmation: confirm responsible user roles, validation rules, approval workflow, notifications, reporting needs and acceptance scenarios for every selected feature.

Procurement & Vendor Management

Feature ID	Feature / capability	Priority
F-311-01	Vendor onboarding and master	Must
F-311-02	Purchase requisition	Must
F-311-03	RFQ creation	Must
F-311-04	Vendor quotation comparison	Must
F-311-05	Purchase order lifecycle	Must
F-311-06	Purchase order approval	Should
F-311-07	Goods receipt note	Should
F-311-08	Partial receiving and backorders	Should
F-311-09	Purchase returns	Should
F-311-10	Supplier performance records	Should
F-311-11	Contracted pricing and validity	Should
F-311-12	Three-way matching checks	Could
F-311-13	Purchase budget limit validation	Could
F-311-14	Procurement history report	Could

Requirement confirmation: confirm responsible user roles, validation rules, approval workflow, notifications, reporting needs and acceptance scenarios for every selected feature.

Invoicing & Accounts Receivable

Feature ID	Feature / capability	Priority
F-312-01	Customer invoice creation	Must
F-312-02	Draft/review/issue/void lifecycle	Must
F-312-03	Line item tax and discount rules	Must
F-312-04	Invoice numbering sequence	Must
F-312-05	Credit note and refund record	Must
F-312-06	Due date and payment terms	Should
F-312-07	Partial and full payment allocation	Should
F-312-08	Overdue invoice reminders	Should

F-312-09	Customer balance and aging	Should
F-312-10	Invoice PDF download	Should
F-312-11	Invoice email dispatch	Should
F-312-12	Invoice export	Could
F-312-13	Invoice adjustment authorization	Could
F-312-14	Financial period lock controls	Could

Requirement confirmation: confirm responsible user roles, validation rules, approval workflow, notifications, reporting needs and acceptance scenarios for every selected feature.

Expenses, Budget & Payroll

Feature ID	Feature / capability	Priority
F-313-01	Employee expense claim	Must
F-313-02	Receipt uploads	Must
F-313-03	Expense categories and policies	Must
F-313-04	Multi-step expense approval	Must
F-313-05	Reimbursements status	Must
F-313-06	Department budgets and spending limits	Should
F-313-07	Budget versus actual summary	Should
F-313-08	Salary structures and components	Should
F-313-09	Payroll period calculation	Should
F-313-10	Earnings and deductions	Should
F-313-11	Payroll review and approval	Should
F-313-12	Payslip generation	Could
F-313-13	Payroll corrections and reruns	Could
F-313-14	Salary privacy controls	Could
F-313-15	Payroll export for authorized finance team	Could
F-313-16	Statutory rule configuration by jurisdiction	Could

Requirement confirmation: confirm responsible user roles, validation rules, approval workflow, notifications, reporting needs and acceptance scenarios for every selected feature.

5.4 Phase 4 Feature Specifications

Configurable Approval & Workflow Engine

Feature ID	Feature / capability	Priority
F-414-01	Workflow template builder	Must
F-414-02	Workflow states and transitions	Must
F-414-03	Conditional routing rules	Must
F-414-04	Sequential approval levels	Must
F-414-05	Parallel approvals	Must
F-414-06	Delegated approvals	Should
F-414-07	Escalation policies	Should
F-414-08	SLA timers and reminders	Should
F-414-09	Approve/reject/request changes	Should
F-414-10	Workflow versioning	Should
F-414-11	Prevent self-approval rules	Should
F-414-12	Approval history and audit	Could
F-414-13	Workflow dry-run validation	Could
F-414-14	Custom form field mapping	Could
F-414-15	Workflow cancellation and rollback policy	Could

Requirement confirmation: confirm responsible user roles, validation rules, approval workflow, notifications, reporting needs and acceptance scenarios for every selected feature.

Notifications & Internal Collaboration

Feature ID	Feature / capability	Priority

F-415-01	In-app notification inbox	Must
F-415-02	Read/unread status	Must
F-415-03	Email notification templates	Must
F-415-04	Event-driven notification triggers	Must
F-415-05	Notification preference controls	Must
F-415-06	Mention notifications	Should
F-415-07	Approval alerts	Should
F-415-08	Due-date reminders	Should
F-415-09	Real-time activity updates via WebSocket	Should
F-415-10	Notification delivery retry	Should
F-415-11	Digest notifications	Should
F-415-12	Team announcement board	Could
F-415-13	Comment thread and reactions	Could
F-415-14	Notification delivery logs	Could

Requirement confirmation: confirm responsible user roles, validation rules, approval workflow, notifications, reporting needs and acceptance scenarios for every selected feature.

Documents & Knowledge Records

Feature ID	Feature / capability	Priority
F-416-01	File upload/download	Must
F-416-02	Object storage integration	Must
F-416-03	File type and size validation	Must
F-416-04	Tenant-scoped file access	Must
F-416-05	Document categories and tags	Must
F-416-06	Document metadata and search	Should
F-416-07	File version history	Should
F-416-08	Folder and document permissions	Should
F-416-09	Expiring access links	Should
F-416-10	Document preview	Should
F-416-11	Retention and deletion policies	Should
F-416-12	Document review and approval	Could
F-416-13	Virus scanning integration	Could
F-416-14	File access audit trail	Could

Requirement confirmation: confirm responsible user roles, validation rules, approval workflow, notifications, reporting needs and acceptance scenarios for every selected feature.

Reports & Analytics (Non-AI)

Feature ID	Feature / capability	Priority
F-417-01	Dashboard widget configuration	Must
F-417-02	KPI cards by role	Must
F-417-03	Employee attendance reports	Must
F-417-04	Leave utilization reports	Must
F-417-05	Payroll summary reports	Must
F-417-06	Project delivery reports	Should
F-417-07	Task aging and throughput	Should
F-417-08	CRM pipeline reports	Should
F-417-09	Deal conversion reports	Should
F-417-10	Inventory valuation reports	Should
F-417-11	Procurement spend reports	Should
F-417-12	Expense and budget reports	Could
F-417-13	AR aging reports	Could
F-417-14	CSV/XLSX/PDF export	Could
F-417-15	Scheduled report generation	Could
F-417-16	Permission-aware report filters	Could
F-417-17	Dashboard data refresh controls	Could
F-417-18	Date-range and branch filters	Could

Requirement confirmation: confirm responsible user roles, validation rules, approval workflow, notifications, reporting needs and acceptance scenarios for every selected feature.

Integration API & Webhooks

Feature ID	Feature / capability	Priority
F-418-01	Versioned REST API	Must
F-418-02	OpenAPI documentation	Must
F-418-03	Client API credentials	Must
F-418-04	Scope-limited integration tokens	Must
F-418-05	Webhook event subscriptions	Must
F-418-06	Webhook HMAC signing	Should
F-418-07	Retry and dead-letter handling	Should
F-418-08	Webhook delivery logs	Should
F-418-09	Import/export jobs	Should
F-418-10	Integration request rate limiting	Should
F-418-11	Idempotency keys for financial writes	Should
F-418-12	API deprecation strategy	Could

Requirement confirmation: confirm responsible user roles, validation rules, approval workflow, notifications, reporting needs and acceptance scenarios for every selected feature.

5.5 Phase 5 Feature Specifications

Platform Subscription & Billing

Feature ID	Feature / capability	Priority
F-519-01	Plan catalog and limits	Must
F-519-02	Trial account policy	Must
F-519-03	Plan upgrade/downgrade	Must
F-519-04	Subscription renewal dates	Must
F-519-05	Seat-based limit enforcement	Must
F-519-06	Usage quota dashboard	Should
F-519-07	Billing contact management	Should
F-519-08	Platform invoice generation	Should
F-519-09	Failed payment handling	Should
F-519-10	Payment gateway adapter (optional)	Should
F-519-11	Grace period and suspension policies	Should
F-519-12	Subscription events audit	Could
F-519-13	Tenant data export on cancellation	Could

Requirement confirmation: confirm responsible user roles, validation rules, approval workflow, notifications, reporting needs and acceptance scenarios for every selected feature.

Security, Privacy & Compliance

Feature ID	Feature / capability	Priority
F-520-01	Threat model and security review	Must
F-520-02	Input validation and output encoding	Must
F-520-03	CSRF and CORS configuration	Must
F-520-04	Secure cookies and TLS	Must
F-520-05	Encryption at rest strategy	Must
F-520-06	Secret management and rotation	Should
F-520-07	Tenant isolation automated tests	Should
F-520-08	Data retention policy	Should
F-520-09	Personal data export/delete requests	Should
F-520-10	Sensitive field masking	Should
F-520-11	Immutable security audit events	Should
F-520-12	Privileged action approval	Could
F-520-13	Backup encryption	Could
F-520-14	Security incident response runbook	Could
F-520-15	Vulnerability and dependency scans	Could
F-520-16	Rate-limit abuse controls	Could

Requirement confirmation: confirm responsible user roles, validation rules, approval workflow, notifications, reporting needs and acceptance scenarios for every selected feature.

Operations, Reliability & Delivery

Feature ID	Feature / capability	Priority
F-521-01	Dockerized environments	Must
F-521-02	CI build/test pipeline	Must
F-521-03	Database migrations	Must
F-521-04	Dev/staging/production configuration	Must
F-521-05	Environment secrets	Must
F-521-06	Centralized structured logging	Should
F-521-07	Metrics and health endpoints	Should
F-521-08	Alert dashboards	Should
F-521-09	Backup automation and restore drills	Should
F-521-10	Disaster recovery runbook	Should
F-521-11	Job queue monitoring	Should
F-521-12	Load and stress testing	Could
F-521-13	Performance budgets	Could
F-521-14	Rollback and release strategy	Could
F-521-15	Feature flags	Could
F-521-16	Production deployment runbook	Could
F-521-17	Admin operations handbook	Could
F-521-18	End-user training materials	Could
F-521-19	UAT and formal sign-off	Could

Requirement confirmation: confirm responsible user roles, validation rules, approval workflow, notifications, reporting needs and acceptance scenarios for every selected feature.

6. Cross-Module End-to-End Scenarios

- Hire to payroll: Onboard employee → assign manager/shift → record attendance and leave → calculate payroll → approve payslip
- Lead to cash: Register lead → convert opportunity → quote → approved order/invoice → record partial/full payment → aging report
- Procure to stock: Requisition → approval → RFQ → purchase order → goods receipt → stock ledger update → invoice matching
- Project execution: Create project → milestone/sprint → assign tasks → log effort → approve timesheets → progress report
- Expense settlement: Claim + receipt → policy validation → manager/finance approval → reimbursement record → audit trail
- Tenant lifecycle: Register company → verify → provision owner → invite users → choose plan → configure org → export or close account

7. Non-Functional Requirements (Client Decisions)

Area	Specification to agree
Security	Tenant isolation, RBAC, MFA scope, encryption, audit retention, app security testing
Reliability	Availability target, RPO/RTO, backup frequency, restore ownership
Performance	Expected simultaneous users, requests/sec, p95 latency, peak payroll/import sizes
Scalability	Tenant count, users per tenant, storage volume and file upload limits
Compliance	Data jurisdiction, statutory payroll/tax rules, records retention and consent
Accessibility	Keyboard operation, contrast, WCAG target, mobile responsiveness
Interoperability	Import sources, API consumers, webhook consumers and

	integrations
Operations	Monitoring, on-call owner, incident severity, maintenance windows

8. Data Model - Main Entity Groups

Domain	Candidate MySQL entities
Platform	tenants, tenant_settings, plans, subscriptions, billing_accounts
Access	users, invitations, sessions, roles, permissions, role_assignments, audit_events
HRMS	employees, departments, branches, attendance, shifts, leaves, salary_structures, payroll_runs
Projects	projects, project_members, sprints, tasks, task_dependencies, timesheets
CRM	leads, contacts, accounts, deals, quotes, contracts
Supply chain	products, warehouses, stock_movements, vendors, requisitions, purchase_orders, goods_receipts
Finance	invoices, invoice_lines, payments, credit_notes, expense_claims, budgets
Platform operations	workflow_templates, approvals, notifications, documents, report_jobs, webhooks

Database design guidance: enforce tenant_id where applicable, use composite tenant-aware indexes and unique constraints, soft-deletion selectively, transactional outbox for reliable event publishing, and explicit row ownership checks. Confirm tax and payroll jurisdiction before detailed schema freeze.

9. Proposed API Surface

API area	Illustrative endpoint groups
Identity	/api/v1/auth/*, /api/v1/sessions/*
Tenancy & users	/api/v1/tenants/*, /api/v1/users/*, /api/v1/roles/*
HRMS	/api/v1/employees/*, /api/v1/attendance/*, /api/v1/leaves/*, /api/v1/payroll/*
Projects	/api/v1/projects/*, /api/v1/tasks/*, /api/v1/timesheets/*
Business	/api/v1/leads/*, /api/v1/deals/*, /api/v1/inventory/*, /api/v1/purchase-orders/*
Finance	/api/v1/invoices/*, /api/v1/expenses/*, /api/v1/budgets/*
Platform	/api/v1/workflows/*, /api/v1/notifications/*, /api/v1/reports/*, /api/v1/webhooks/*

These are preliminary route groups, not final OpenAPI contracts. Define HTTP semantics, request/response DTOs, pagination, authorization, validation errors and idempotency per confirmed use case.

10. Minimum Acceptance Criteria

- No user can read or modify records belonging to another tenant through UI, API, background jobs or file links.
- Every privileged mutation is permission checked and generates an appropriately scoped audit event.
- Financial and inventory postings maintain consistent ledger balances across retries and concurrent requests.
- Critical business paths are covered by automated tests and signed-off UAT scenarios.
- Exports, reports and notifications respect tenant boundaries and permission filters.
- Restore drills demonstrate the agreed backup/recovery objective before production acceptance.

11. Client Feature Selection & Sign-off Worksheet

For every feature, client should indicate: Approved / Deferred / Rejected; Priority; Owner; Acceptance Criteria; Legal/Policy Constraints; Integration Dependency; Target Release. The accompanying CSV can be edited as a backlog and imported into project tools.

Approval item	Client decision
Business owner / sponsor	_____
Target organization(s)	_____
Deployment model	Cloud / On-premises / Hybrid
Selected MVP modules	_____
Statutory jurisdiction	_____
Security/compliance requirements	_____
Client name and date	_____
Vendor/technical lead sign-off	_____

Change control: any changes to signed baseline scope should include estimated impact on timeline, cost, dependencies, testing and documentation before approval.